The DICTO Hecatogdoecontahedron

The DICTO Hecatodihedron and the DICTO Ogdoecontahedron interlocked as a self-dual compound of 182 faces

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Licence CC BY 4.0 · An introduction to one solid; its parent paper: Progenitors, DICTO DOI: 10.5281/zenodo.23413184 (and The DICTO Truncated-Icosahedral Hecatodihedron, Version 2, DICTO DOI: 10.5281/zenodo.23360857). Data and checks are included with this record.
The pairing and the name are DICTO's; the exact computations and the text were done with an AI assistant (Claude, Anthropic).

Abstract. The DICTO Hecatodihedron H (102 faces, 80 corners) and its polar, the DICTO Ogdoecontahedron O (80 faces, 102 corners), interlock in the truncated icosahedron's midsphere as one compound: the DICTO Hecatogdoecontahedron (the two names merged), 182 faces and 182 corners, self-dual. Its common core is the rectified truncated icosahedron and its hull the joined truncated icosahedron (the rhombic enneacontahedron), the core and hull of the classical truncated icosahedron–pentakis dodecahedron compound. We claim the pairing and the name only, not the core or the hull. Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float.

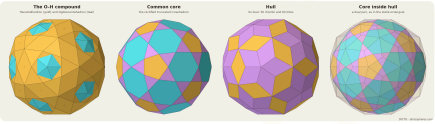


Figure 1. The DICTO Hecatogdoecontahedron: H (gold) and O (teal) interlocked, with the truncated icosahedron and the common core and hull. Drawn from exact corners.

1. The compound

Take the Hecatodihedron H of the Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833) and reciprocate it in the sphere of squared radius 9/4, the midsphere of the Dharma's truncated icosahedron, which touches H's 90 rhombi and triangles. The polar is the Ogdoecontahedron O at its compound size. Then (exact): the 90 corners of O on that sphere are exactly the 90 edge midpoints of the truncated icosahedron; no corner of H lies inside O and no corner of O inside H.

Part	Corners	Faces
H, the DICTO Hecatodihedron	80	102: 12 pentagons, 30 rhombi, 60 triangles
O, the DICTO Ogdoecontahedron	102	80: 20 regular hexagons, 60 kites
the compound	**182**	**182**; self-dual (a solid with its own polar)
common core H ∩ O	90	92: the rectified truncated icosahedron (12 regular pentagons, 20 regular hexagons, 60 triangles), all corners on the midsphere
hull	92	90 rhombi: the joined truncated icosahedron (rhombic enneacontahedron), the core's polar

Table 1. The DICTO Hecatogdoecontahedron (`oh_compound.py`; corner and face counts exact, volumes float).

2. Where it appears

The compound was first described in Version 2 of the Hecatodihedron paper (DICTO DOI: 10.5281/zenodo.23360857). In Progenitors (DICTO DOI: 10.5281/zenodo.23413184) its core and hull, rTI and jtI, become stations of the golden cycle I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ I·φ^3, so the cycle holds the core and hull of two dual compounds: the classical icosahedron–dodecahedron one (ID, RT) and this one. As the octahedron and cube are the core and hull of the stella octangula, the rectified truncated icosahedron and the rhombic enneacontahedron are those of the Hecatogdoecontahedron.

3. Status

Known: compounds of a solid with its polar; the truncated icosahedron–pentakis dodecahedron compound, the rectified truncated icosahedron and the rhombic enneacontahedron (MathWorld; Wikipedia; McCooey; Hart). **Not found, in the sources checked** (Progenitors §4, SEARCH-LOG.md): the compound of H and O. The name returns no results. Every "not found" is limited to those sources.

Sources checked on the web (17 October 2026):

- **MathWorld**, "Truncated Icosahedron–Pentakis Dodecahedron Compound": the classical compound; no core, hull or H–O compound.
- **Wikipedia**, "Rectified truncated icosahedron" (its dual is topologically the rhombic enneacontahedron) and "Truncated icosahedron".
- **McCooey, Visual Polyhedra**: Rectified and Truncated Archimedean Solids; no compounds are listed.
- **G. Hart, Virtual Polyhedra**: the compound list and the full alphabetic model list (read for Progenitors): compounds of the uniform solids with their duals; not this one.
- **Web search** for the name: no results.

Polyhedron software libraries (17 October 2026):

- **SageMath** polytope library (`polytopes.*`): its icosahedral constructors are the icosahedron, dodecahedron, icosidodecahedron, truncated icosahedron and the other Archimedean solids, and the pentakis dodecahedron; none of the solids here.
- **netlib polyhedra database** (142 entries: Platonic, Archimedean and their duals, prisms, the 92 Johnson solids and five others): none of the solids here (the pentakis dodecahedron, file 41, is the canonical one).
- **Antiprism** 0.29 (A. Rossiter): used to recompute and check the structures independently (for Vajrayana and Progenitors, and the companion's type t3jD); its Conway-operator sweep is a generator, not a catalogue.
- **Not run:** Wolfram `PolyhedronData` (needs Mathematica) and Stella / Great Stella (R. Webb; its library beyond the uniform, Johnson and Stewart-toroid models is not listed online). No claim is made about them.

The reference library: see the Appendix.

4. Files

In the study (branch research: studies/2026-10-16-progenitors): `oh_compound.py` → `oh-compound.txt`, `core_hull_in_cycle.py`. See it in the app: Kaleidohedra, 3D+ → Vajrayana (its core

and hull are stations) and Multi-shell chains.

Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md) holds 234 entries. The full text of the 64 entries below was searched on 17 October 2026 for the solid's construction, its key numbers and its name (pentakis / kis-dodecahedron / hexecontahedron / hull of a dodecahedron and an icosahedron; enneacontahedron / rectified truncated icosahedron / compounds of the truncated icosahedron; chamfered or truncated triacontahedra; the four DICTO names). None contains any of the four solids introduced in these papers; the only hits are mentions of the classical pentakis dodecahedron (Kepler 1619; Messer 1995; de Graaf, van Roij and Dijkstra 2011) and of the chamfer operation (Goldberg 1937; the Goldberg-polyhedron and Fullerene articles). Not text-searched: 12 image-only scans and 158 link-only entries (copyrighted books and articles held as citations); no claim is made about them.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1619	Harmonices mundi libri V
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional en
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric

K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
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C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron